Unit 2 Quiz: Analyzing One Variable

Name _____

Show your work.

2.1 - Standard Deviation

1. **I find the standard deviation of a small data set.**

   Kim counted how many text messages she got each hour for 5 hours.

   2, 6, 0, 4, 3

   (a) Find the mean, $\bar{x}$. Then complete the table.

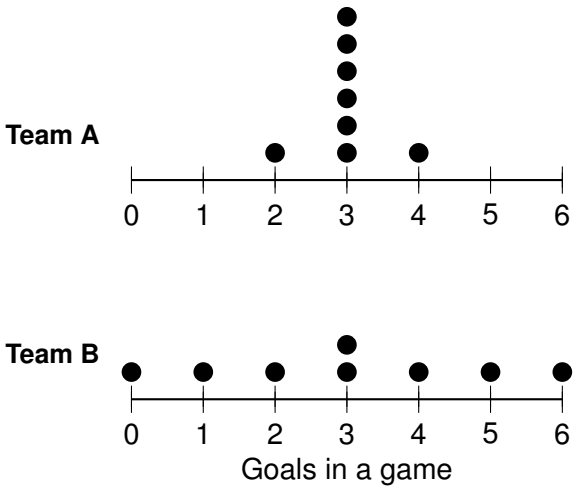
   $\bar{x}$ = _____

x	$x - \bar{x}$	$(x - \bar{x})^2$
2		
6		
0		
4		
3		
	$\Sigma(x - \bar{x})^2$	

   (b) Find the standard deviation, s. Round to 2 decimal places.

2. **I compare the spread of data sets by looking at graphs, and I explain what the standard deviation tells about data.**

   These dot plots show the goals two soccer teams scored in 8 games. Both teams have a mean of 3 goals.



   (a) Which team has the larger standard deviation? Explain.

   (b) Team B's SD is 2 goals. What does this number tell you about Team B?

   (c) Team C scored exactly 3 goals in every game. What is Team C's SD? Explain.

2.2 - Choosing Measures and Comparing Distributions

3. **I choose mean and SD or median and IQR based on the shape of the data.**

   For each data set, circle the measures you should use. Then explain.

   (a) The minutes 30 students spent on the bus today. The histogram is skewed right.

   mean and SD median and IQR Explain:

   (b) The lengths of 100 leaves from one tree. The histogram is symmetric, with no outliers.

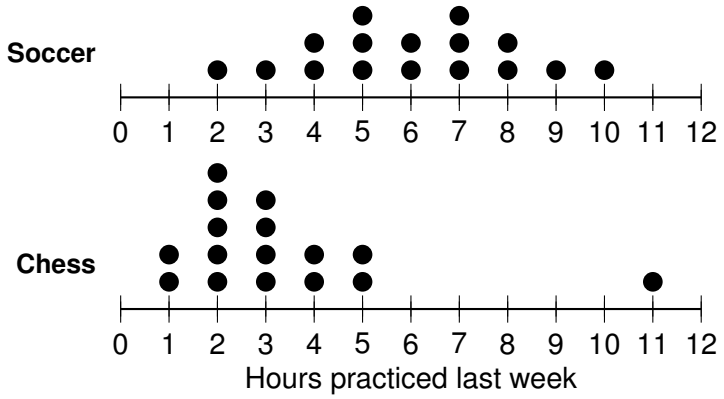
   mean and SD median and IQR Explain:

   (c) The prices of 7 used phones, in dollars: 120, 140, 150, 150, 160, 180, 450

   mean and SD median and IQR Explain:

4. **I compare two distributions using shape, center, spread, and outliers.**

   These dot plots show how many hours students in two clubs practiced last week.



	Q1	Median	Q3	IQR
Soccer	4.5	6	7.5	3
Chess	2	3	4	2

   (a) Describe the shape of each club's data.

   (b) Compare the centers. Use numbers and the context.

   (c) Compare the spreads. Use numbers.

   (d) Does either club have an outlier? Explain.

2.3 - Percentiles and Z-Scores

5. **I find the percentile of a value.**

   20 students said how many minutes they played a phone game last night. Here are the times, in order.

   5, 8, 12, 15, 18, 20, 22, 25, 27, 30, 32, 35, 38, 40, 42, 45, 50, 55, 60, 75

   (a) What percentile is 45 minutes?

   (b) What percentile is 15 minutes?

   (c) Which time is at the 40th percentile?

6. **I find and explain a z-score, and I compare values from different data sets with z-scores.**

   Amina took two tests.

Test	Amina's score	Class mean μ	Class SD σ
Math	79	73	4
Reading	91	83	8

   (a) Find Amina's z-score on each test.

   (b) What does Amina's z-score on the math test tell you?

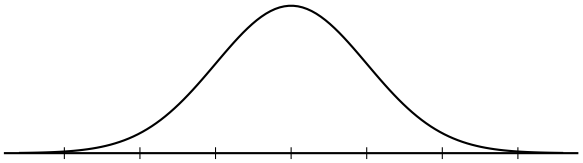
   (c) Amina's reading score is higher. On which test did she do better compared to her class? Explain.

2.4 - Normal Distributions and the Empirical Rule

7. **I use the empirical rule (68-95-99.7) to find percents, and I find the percentile of a value 1, 2, or 3 SDs from the mean.**

The high temperature each day in July in one city has a normal distribution. The mean is 88°F and the SD is 4°F.

(a) Label the number line with the temperatures.



High temperature (°F)

(b) About what percent of days have a high temperature from 84°F to 92°F?

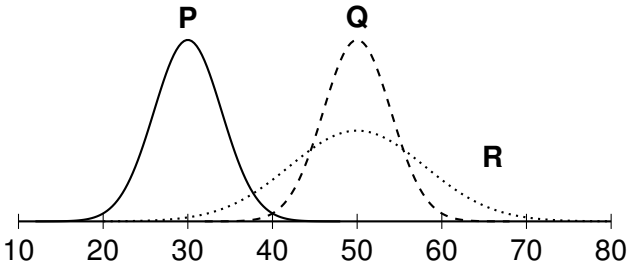
(c) About what percent of days have a high temperature above 96°F?

(d) About what percent of days have a high temperature from 80°F to 92°F?

(e) What percentile is a high temperature of 92°F?

8. **I describe a normal distribution, and I compare normal curves with different means and SDs.**

The three curves are normal distributions.



(a) Which two curves have the same mean?

(b) Which two curves have the same SD?

(c) Which curve has the largest SD? Explain.

(d) Which statement is true for every normal distribution? Circle the letter.

A The mean is greater than the median.

B The mean and the median are equal.

C The data is skewed right.

D The mean is always 0.

2.5 - Normal Calculations with a Z-Table

Use the z-table.

9. **I read a z-table.**

Find the area **below** each z-score.

(a) $z = 0.73$

(b) $z = -1.45$

Find the area **above** or **between**.

(c) above $z = 1.10$

(d) between $z = -1.30$ and $z = 0.40$

10. **I find the percent of data below, above, or between values in a normal distribution.**

A machine fills bottles with water. The amounts have a normal distribution with a mean of 500 mL and an SD of 6 mL. What percent of the bottles have:

(a) less than 510 mL?

(b) more than 491 mL?

(c) from 497 mL to 509 mL?

2.6 - Normal Calculations in Reverse

Use the z-table.

11. **I find the z-score for a percentile with the z-table, and I find the value at a percentile in a normal distribution.**

    (a) Find the z-score at the 60th percentile.

    (b) Find the z-score at the 45th percentile.

    Scores in a video game have a normal distribution. The mean is 1,200 points and the SD is 200 points.

    (c) What score is at the 35th percentile?

12. **I find cutoffs for "top" and "bottom" percents.**

    The weights of a bakery's loaves of bread have a normal distribution. The mean is 680 grams and the SD is 12 grams.

    (a) The lightest 4% of the loaves are sold at a lower price. Below what weight is a loaf sold at a lower price?

    (b) The heaviest 30% of the loaves get a "large" sticker. How heavy must a loaf be to get the sticker?